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RDMA Transports in Datacenter Networks: Survey

Jinbin Hu, Houqiang Shen, Xuchong Liu, and Jin Wang

Abstract

Remote Direct Memory Access (RDMA) has become an important building block of modern datacenter network (DCN) infrastructure given the merits of kernel bypass, zero memory copy techniques and low CPU overhead. However, the increasingly stringent performance requirements of both ultralow latency and high throughput from the booming datacenter applications raise substantial challenges to the widespread adoption of RDMA in large-scale DCNs. This article identifies and discusses specific challenges facing RDMA deployment, including congestion control, multi-path transmission and scalability issues. We provide a comprehensive review of the efforts undertaken over the past decade to enhance RDMA performance, highlighting the key innovations of these RDMA transports. Furthermore, we explore the insights behind these RDMA transport schemes and outline the future research opportunities with the hope to inspire new interests and shed light on the following research in this field.

Introduction

As the datacenter network (DCN) bandwidth continuous to increase, from 10Gbps to 100Gbps or even higher, network transmission becomes the bottleneck for datacenter applications [2], [3]. Specifically, traditional kernel TCP cannot meet the demand of latency-sensitive services like online search or bandwidth-intensive applications like distributed machine learning and cloud storage [6, 11, 15]. In recent years, Remote Direct Memory Access (RDMA) network has become an attractive trend in data centers to meet the above rising stringent demands by providing ultra-low latency, high throughput and low CPU overhead with the kernel bypass and zero memory copy techniques. Nowadays, RDMA networks have been deployed using RDMA over Converged Ethernet Version 2 (RoCEv2) protocol in production data centers such as Microsoft, Google and Alibaba data centers [1, 2, 6, 15].

Specifically, the direct memory-to-memory data transfer in RDMA network allows servers to exchange data in the main memory without involving the processor, cache, or operating system of either server. The typical RDMA data flow is shown in Figure 1. The first step of data transfer process using RDMA is establishing an RDMA connection between two nodes (requester and responder), which involves setting up Queue Pairs (QPs) on both nodes. A QP is essentially a Work Queue (WQ) that holds work tasks, known as Work Requests (WRs). There are two types of WQs: the Send Queue (SQ) is responsible for sending tasks, while the Receive Queue (RQ) handles incoming tasks. The applications package the tasks to be performed by RDMA NIC (RNIC) into Work Queue Elements (WQEs) and post them to the QP. The RNIC retrieves these WQEs from the QP and completes the tasks as defined by the WQEs. After the RNIC completes a WR, it packages the completion information into a Completion Queue Element (CQE) and places it into the Completion Queues (CQs). The applications then poll CQs to retrieve CQEs, which informs them that the RNIC has completed a specific WR. In brief, RDMA operations can be performed directly from user-space applications, bypassing kernel intervention and without intermediate copying by the CPU, thus reducing context-switching overheads.

However, RDMA networks encounter several fundamental challenges during the process of actual deployment and operation [4]. If deploying RDMA directly in Ethernet-based DCNs, RDMA throughput degrades dramatically, because only the simple go-back-N retransmission mechanism is implemented in the RDMA NICs (RNICs) due to limited on-chip resources [6]. Consequently, hop-by-hop Priority-based Flow Control (PFC) is used to enable lossless RDMA networks by avoiding buffer overflow at the switches. However, PFC brings new issues including head-of-line blocking, congestion spreading and deadlock, decreasing RDMA's performance and hurting RDMA's scalability [2, 11, 12]. Recently, efficient and scalable lossy RDMA has attracted extensive attention in academia and industry while redesigning RDMA protocol becomes the new trend [15].

In the light of the above problems in lossless and lossy RDMA networks, many research studies have been working on how to improve RDMA transmission performance in DCNs. Figure 2 shows the timeline of typical RDMA transport solutions in the past decade. Since RoCEv2 [1] was proposed in 2014, a series of excellent related works have been published one after another. Table 1 compares the properties of existing RDMA transports in terms of single-path/multi-path topology, lossless/lossy network dependency, congestion control method and implementing location. DCQCN [2], TIMELY [3], IRN [7], HPCC [8], Swift [10], PCN [11] and

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PACC [13] focus on designing single-path congestion control mechanism to improve application performance. They are end-to-end transport protocols, which detect congestion at the switch and then notify the congestion to end-hosts to adjust sending rate for mitigating congestion. Except for IRN [7], which works in lossy RDMA networks with improved packet retransmission mechanism, all other protocols work in lossless RDMA networks. MP-RDMA [6], PLB [14] and Proteus [17] are proposed to make full use of multiple equal-cost paths and alleviate congestion by fine-grained load balancing. ConWeave [16] focuses on avoid out-of-order packets in network for existing load balancing mechanisms. They are deployed at switch for flexible re routing. Tagger [5], GFC [9] and TCD [12] are helpful to address specific PFC issues such as head-of-line blocking, PFC storms and deadlock revealed in [4]. The latest work SRNIC [15] has taken a step towards improving RDMA scalability by designing scalable RDMA NIC.

In this article, we conduct a comprehensive survey of RDMA transport techniques from the perspectives of singlepath congestion control, multi-path transport, and miscellaneous issues. We first present the background and motivations for each research topic, and then discuss their research challenges. Next, we introduce the state-of-the-art RDMA transport solutions and discuss the future research directions for RDMA transports in Ethernet-based DCNs. Finally, we conclude the article.

Single-Path Congestion Control

Background and Motivations

With the link speed growing rapidly from 10Gbps to 100Gbps or higher in DCNs, new flows starting at line rate will aggressively seize the available bandwidth, leading to deep queueing buildup [15]. Moreover, with the datacenter traffic bursty strength increasing continuously, especially in the Partition/Aggregate communication patterns, a bulk of in-flight packets transmitted simultaneously from a large number of distributed source end-hosts to a small number of destinations are easy to cause serious transient congestion.

Given that RoCEv2 protocol offloaded on the hardware-based RDMA NIC lacks congestion control mechanism, how to detect congestion and adjust sending rate in coordination with RoCEv2 is a crucial research topic. In addition, limited by the hardware resources of the RDMA NIC, the packet loss recovery mechanism in RoCEv2 implements the simple go-back-N strategy, which retransmits all the packets that have not received acknowledge (ACK) packets after the lost packet. The inefficient retransmission causes RDMA to be very sensitive to packet loss. Research shows that RDMA throughput degrades dramatically to nearly zero when the packet loss rate exceeds 0.1% [6], which also makes congestion control particularly important.

Without timely and effective congestion control, the application performances in both lossless and lossy RDMA networks will be seriously damaged due to large tail latency caused by queueing. One line of research concerns the PFC-enabled lossless DCNs [2, 3, 8, 10, 11], once the ingress queue length exceeds a specified threshold, PFC mechanism is triggered to pause the upstream port or a priority class to prevent buffer overflow. This coarse-grained flow control brings a series of negative problems such as the well-known head-of-line blocking, congestion spreading and even extremely serious PFC storms, deadlock, etc. The authors in DCQCN [2] show that PFC severely damages the performance of RoCEv2 in terms of fairness and innocent flows. For example, the median throughput of victim flow falls from 10Gbps to

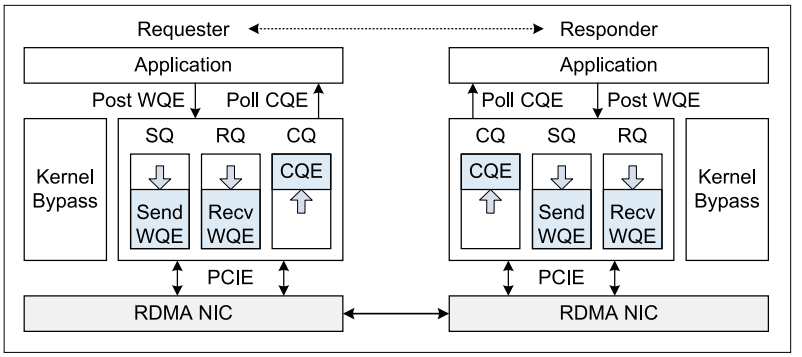


FIGURE 1. Typical RDMA data flow.

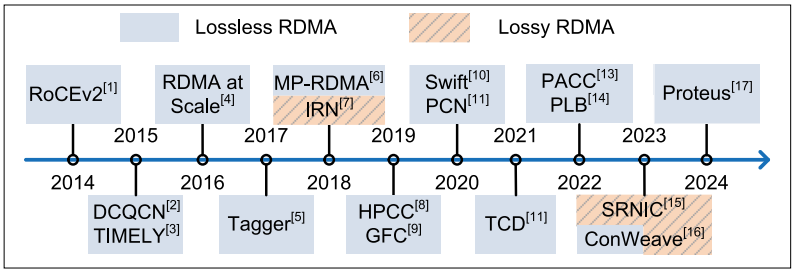


FIGURE 2. Timeline of typical RDMA transport solutions in the past decade.

Scheme	Topology	Reliability	Congestion Control	Implementing Location
DCQCN [2]	single-path	lossless	rate adjusting	switch,NIC
TIMELY [3]	single-path	lossless	rate adjusting	switch,NIC
IRN [7]	single-path	lossy	rate adjusting	switch,NIC
HPCC [8]	single-path	lossless	rate adjusting	switch,NIC
Swift [10]	single-path	lossless	rate adjusting	switch,NIC
PCN [11]	single-path	lossless	rate adjusting	switch,NIC
PACC [13]	single-path	lossless	rate adjusting	switch,NIC
MP-RDMA	multi-path	lossless	rate adjusting & packet rerouting	switch,NIC
ConWeave	multi-path	lossless /	packet rerouting	lossy
PLB [14]	multi-path	lossless	packet rerouting	switch
Proteus [17]	multi-path	lossless	packet rerouting	switch
TCD [12]	-	lossless	-	switch
Tagger [5]	-	lossless	-	switch
GFC [9]	-	lossless	rate adjusting	switch
SRNIC [15]	-	lossy	-	NIC

TABLE 1. Comparison of properties of RDMA transports.

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4.5Gbps due to the head-of-line blocking problem caused by PFC PAUSE frames cascading diffusion. Another line of research focuses on the lossy RDMA networks [7], [15], a lot of lost packets due to congestion will trigger many unnecessary retransmission or even timeouts, leading to large tail latency. As stated in IRN [7], the performance of RoCE without PFC decreases by 1.5 to 3 times due to the increased congestion and the wasted bandwidth caused by redundant retransmission packets. Therefore, congestion control is critical to the system performance in data centers.

Challenges

Congestion control has always been a crucial way to improve the datacenter application performance and user experience. However, effective congestion control is not easy to achieve, especially in modern DCNs with increasing link bandwidth. The key challenges are the timeliness and accuracy of congestion control.

The first challenge is timely control congestion. The measurement studies have shown that more than 60% of the flows in realistic datacenter workloads can be finished just in one RTT or a few RTTs [14]. It is difficult to control these tiny flows in an end-to-end manner. Although the above existing single-path congestion control protocols effectively alleviate persistent congestion, they are not able to handle transient congestion well. The reason is that these end-to-end solutions require at least one RTT control loop from congestion sensing to sending rate adjustment based on the congestion feedback. Therefore, how to improve tolerance for bursty traffic, mitigate the instantaneous congestion in time, shorten the period from congestion detection to congestion control taking effects, and eliminate PFC triggering in lossless networks or reduce congestion losses in lossy networks are extremely challenging research topics.

The second challenge is accurately control congestion. First, it is not trivial to choose appropriate signals to convey congestion correctly. At present, congestion signals such as ECN, RTT and INT have their own pros and cons. Specifically, ECN is binary signal that is marked as long as the queue length of one hop exceeds a preset threshold. Thus, ECN can only indicate that there is congestion on the path, but cannot reflect the congestion degree at each local switch. RTT variation also makes it difficult to distinguish whether the delay is caused by in-network congestion or processing delay at the end-hosts. INT technology introduces non-negligible additional traffic overhead. Therefore, accurate congestion detection requires to take multiple signals indicating congestion into consideration. Second, distinguishing the congested flows that really responsible for the congestion is quite challenging due to multiple congestion bottlenecks. Third, performing accurate rate adjustment for the congested flows is also quite challenging. Existing solutions generally require multiple RTTs for rate evolution to converge to the target sending rate accurately.

In brief, it is quite challenging to detect and control congestion timely and accurately in RDMA DCNs. Especially under the bursty congestion scenarios, the rate adjustment of end-to-end congestion control is difficult to match with the fast operations of hop-by-hop PFC in the current mainstream lossless networks.

Existing Solutions

To enable RDMA networks operate effectively, many research efforts have been spent on designing efficient congestion control to reduce PFC triggering or packet dropping in lossless or lossy RDMA networks in the past few years.

For Lossless RDMA Networks. As the earliest representative RDMA congestion control protocol, DCQCN [2] leverages Explicit Congestion Notification (ECN) marked packets to indicate queue length at the switches and uses Congestion Notification Packets (CNP) generated at the receiver to inform senders to adjust sending rate. DCQCN has been implemented in Mellanox ConnectX series NIC and is being deployed in Microsoft's data centers. Since DCQCN uses a fine-grained congestion control to avoid triggering PFC persistently, it effectively cures PFC's ills and addresses the tussle between responsiveness and stability. For example, with incast degree of 10, the 10th percentile throughput with DCQCN is increased by more than twice compared to that without DCQCN.

TIMELY [3] and its evolution protocol Swift [10] are deployed in Google data centers, they use the Round Trip Time (RTT) variation measured in the hardware NIC with microsecond accuracy to predict congestion and correspondingly control the sending rate at the end-hosts. By utilizing the NIC hardware timestamps and implementing a rapid response mechanism to congestion in the protocol stack, Swift achieves excellent performance in line-rate throughput with low latency and almost zero loss under the 5000 -to-1 incast scenario.

HPCC [8] is presented as a high precision congestion control, it employs in-network telemetry (INT) technology supported by new switching Application Specific Integrated Circuit (ASIC) to obtain precise link utilization and compute accurate flow rate without gradual evolution and convergence process. HPCC keeps a near-zero queue with no packet loss and no PFC triggering, and therefore the flow completion time (FCT) of short flow is close to the base RTT, while the ECN-based protocols such as DCQCN keep a standing queue around the ECN marking threshold with the latency consistently higher than 6 times.

However, the above solutions do not recognize the congested flows that really responsible for congestion, they treat all flows passing through the congestion point equally, resulting in unreasonable rate adjustment. To further control traffic precisely, PCN [11] is the first congestion control protocol that identify real congested flows and make discriminative rate adjustment for congested and uncongested flows. Moreover, PCN adopts a receiver-driven rate adjustment scheme to throttle the incoming traffic to match the bottleneck capacity directly, rather than detect the available bandwidth round by round. Thus, PCN can limit the bottleneck queue length at a very low level of a few packets in no more than 15 RTT and regulate the aggregate sending rate to the bottleneck capacity within 4 RTT, which is 25 times and 45 times faster than that with DCQCN and TIMELY, respectively.

For Lossy RDMA Networks. Unlike the above congestion control mechanisms, which assume

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working in a lossless networking fabric, the recently proposed IRN [7] operates in the lossy RDMA networks without enabling PFC and its negative impacts. IRN redesigns a more efficient packet loss recovery mechanism and bounds the number of in-flight packets for each flow by the bandwidth-delay product (BDP) of the network on the RoCE NIC with low overhead. Since IRN without PFC can recover from single-packet message losses quickly due to the low timeout value, IRN always performs better than RoCE with PFC by up to 83% across various cases. The improved packet selective retransmission and packet level flow control method make RDMA more resilient to the large-scale lossy DCNs. However, the performance in terms of convergence and fairness needs further exploration and improvement.

Future Opportunities

Based on the limitations of current single-path congestion control methods, we identify the potential of synthesizing multiple signals for congestion detection, resilient congestion control mechanisms and hardware-based transport protocols for enhancing congestion control in RDMA networks. We draw upon a comprehensive strategy that encompasses the synthesis of diverse signal information, the utilization of advanced NICs, and the integration of in-network programmable switching ASICs for refined congestion detection and control.

First, based on the respective advantages of various signals that can piggyback more different information, synthesizing multiple signals to make them complementary and jointly indicate congestion is a feasible congestion detection mechanism. Implementing this strategy involves developing algorithms that can aggregate and analyze multiple signals (e.g., packet loss rates, latency metrics, bandwidth utilization) to form a coherent assessment of network congestion. In addition, supported by the advanced NIC and programmable switching ASICs, combining multiple congestion feedback points with collaborative notification to distinguish real congested flows and locations where congestion actually occurs in a finegrained manner is a possible method worth trying.

Second, the existing end-to-end congestion control protocols have not yet solved the challenges of controlling bursty congestion. However, in practice, a large number of small flows send all the data before the congestion control schemes take effects. Therefore, designing new resilient congestion control protocols collaboration with in-network devices and end-hosts to alleviate transient congestion due to bursty tiny flows is an important direction. Thanks to the vigorous development of machine learning methods and strong support of computing resources, adaptive adjusting the congestion control method or tuning congestion parameters for time-varying network environments and specific traffic characteristics generated by various applications is also a possible direction. It is a good opportunity to develop protocols that dynamically adjust to network conditions, integrating machine learning algorithms to predict congestion and adjust control parameters in real-time, using feedback from both in-network devices and end-hosts to manage congestion proactively.

The existing single-path transports fall short to take advantage of rich network capacities.

Third, based on the hardware offloading technologies and sufficient on-chip computing and storage resources of the hardware such as Field Programmable Gate Array (FPGA), it is possible to design better hardware-based transport protocols to control congestion with low CPU overhead. Thus, how to leverage hardware offloading and on-chip resources to handle congestion control tasks and reduce reliance on the CPU is challenging, how to design and implement transport protocols on FPGA platforms, optimizing them for efficient congestion management with minimal CPU overhead is also a possible future research direction.

To validate the above approaches, it is necessary to conduct simulations and real-world network testing. Performance metrics such as accuracy in congestion detection, latency reduction, and throughput improvement serve as benchmarks for evaluating effectiveness.

Multi-Path Transport

Background and Motivations

Modern datacenter networks such as Clos topologies provide massive equal-cost parallel paths between end-hosts pairs [6]. The existing single-path transports fall short to take advantage of rich network capacities. In order to make full use of the multiple paths, researchers have proposed a series of multi-path transports and load balancing mechanisms for DCNs [6, 14, 17]. However, these solutions designed for traditional DCNs have poor performance if they are directly employed in RDMA networks. As revealed in PLB [14] and its extended work Proteus [17], link load cannot reflect PFC pausing correctly, end-to-end signals cannot detect PFC pausing in a timely manner, local queue cannot sense remote PFC pausing, and the performance of existing solutions (e.g., CONGA [18], Hermes [19] and DRILL [20]) decreases by up to 49% compared with the ideal situation.

The specific reasons are as follows: First, aligned with the recent trend of hardware offloading, the entire protocol stack processing sinks to the RDMA NIC. Unfortunately, the hardware on-chip resources of the NIC is limited. If the traditional multi-path transmission protocols such as MPTCP are directly deployed in RDMA networks, they will consume more memory than all the available Static Random Access Memory (SRAM) on the NIC [6]. Because these protocols maintain excessive states for flows and sub-flows, leading to high latency and CPU overhead caused by the additional memory footprint between the host memory and the on-NIC SRAM. Second, the signals sensing path status in traditional load balancing schemes cannot perceive PFC PAUSE/RESUME correctly and timely in the PFC-enabled lossless DCNs. Moreover, the congestion feedback signals are potentially blocked due to PFC pausing. For these reasons, it is likely that the sending rate adjustment and load balancing decisions are incorrect or inaccurate. Third, RDMA networks are more sensitive to lost and

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out-of-order packets. The existing multi-path transports are difficult to ensure that packets transmitted over multiple paths arrive at the receiver in an orderly manner, and they are also difficult to control the degree of disorder. The out-of-order packets are mistaken for packet loss, leading to frequent retransmissions and significantly increasing the flow completion time. In short, the above issues motivate researchers to redesign multi-path transport for RDMA networks.

Challenges

The current trend is to offload RDMA protocols to the hardware NICs. It is quite challenging to design and implement a practical multi-path transports on the NIC to operate efficiently in production DCNs. First, to make congestion-aware load balancing decisions, the multi-path transports require to track the congestion states for each path and maintain the information for all the active flows on the hardware with limited computing and storage resources. These states consume considerable hardware resources overhead, and they increase linearly with the number of parallel paths and active flows. Second, due to the diversity of multiple paths in terms of latency, the packets of the same flow allocated on different parallel paths potentially arrive at the receiver out of the order. Especially in high-speed RDMA networks, the simple hardware-based retransmission caused by out-of-order packets will significantly downgrade throughput. Third, in addition to the challenges encountered by the single-path congestion control in RDMA networks, solving the problems of congestion isolation and congestion mismatch among multiple paths for multi-path transmission is also quite challenging.

Existing Solutions

The existing solutions for multiple transports fall into two main categories, including multi-path congestion control and load balancing mechanisms. As the first representative work of RDMA multi-path protocols, MP-RDMA [6] focuses on utilizing multiple paths with minimal memory footprint between host memory and RDMA NIC memory to achieve high throughput and low latency. MP-RDMA proposes three new techniques to solve the challenges of limited RDMA NIC memory. First, to avoid per-path status, MP-RDMA employs a multi-path ACK-clocking mechanism to balance load among parallel paths in a congestion-aware manner. Second, to control the degree of disorder, MP-RDMA utilizes an out-of-order aware path selection scheme to shield the congested path with large delay and maintains a bitmap with low storage overhead to track the reordering packets. Third, MP-RDMA designs a synchronize mechanism to guarantee in-order memory updating. Based on the finer grained load balance and accurate queue length control for congestion control, MP-RDMA effectively reduces the average FCT by up to 17.7% and greatly improve the robustness under path failures. However, MP-RDMA is also an end-to-end transport, and it is difficult to deal with the transient congestion due to bursty traffic in a timely manner. In addition, although MP-RDMA discards the concept of sub-flow to reduce memory consumption caused by states maintenance, it leads to poor congestion isolation among multiple paths, leading to congestion mismatch and longer convergence process.

As the complementary mechanisms to congestion control schemes, PLB [14] and its extended work Proteus [17] focus on balancing load effectively at the switches in PFC-enabled datacenter networks. Through extensively experiments, PLB and Proteus have pointed out that the existing load balancing solutions for lossy DCNs cannot work well in lossless PFC-enabled RDMA networks, because the congestion signals used to guide rerouting, such as separate ECN, RTT or local queue length, could not accurately feed back PFC pausing and resuming in a timely manner. Thus, PLB and Proteus proposed a PFC-aware load balancing mechanism by using RTT-level and sub-RTT level signals to reflect path status simultaneously, requiring switches modifications. Proteus reduces the average FCT and 99th percentile FCT by up to 35% and 46% respectively under web search realistic workload, especially for short flows, with an average FCT reduction of up to 1.8 times. Since the existing load balancing schemes cannot work well in RDMA network due to the unique traffic flow characteristics and the assumption of packet delivery in sequence, ConWeave [16] proposes a new load balancing framework designed for RDMA. ConWeave performs fine granularity rerouting and avoids out-of-order packets at the programmable switches. Compared to the state-of-the-art load balancing algorithms, ConWeave achieves up to 42.3% and 66.8% improvement for average FCT and 99th percentile FCT, respectively.

Future Opportunities

Modern datacenter networks enable abundant multiple paths to provide bisection bandwidth. It is a promising avenue to improve throughput and minimize flow completion time by making full use of parallel paths. The inspiration for this direction comes from the challenges presented by existing multi-path transmission and the untapped potential of programmable network technologies. Despite the potential benefits, there exists a notable gap in the development of multi-path transports. To the best of our knowledge, there are no multiple transports including congestion control protocols and load balancing mechanisms for lossy RDMA networks, and few solutions for lossless DCNs. Therefore, addressing this gap necessitates to design efficient and reliable multipath transports for both lossless and lossy RDMA networks to improve application performance.

One of the possible direction is multi-path congestion control focusing on adjusting sending rate to an optimal target and allocating packets to the appropriate paths simultaneously. Current multi-path transports employ one congestion window for each flow for all parallel paths. Thus, the sending rate is affected by the congestion states of multiple paths, inevitably causing the current rate to be unable to match the status of all paths. To avoid poor congestion isolation among multiple paths and congestion mismatch between the flow rate and path status, we can design and implement fine-grained congestion control algorithms to reasonably adjust the sending rate and

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dynamically select forwarding path based on real-time congestion degree of each path.

The other possible direction is load balancing focusing on rerouting traffic to alleviate congestion without necessarily adjusting the sending rate. By leveraging the programmable switches, we can flexibly modify the packets forwarding logic at the data plane for selecting appropriate egress port. Fully considering the path diversities, we can research how to avoid out-of-order packets to guarantee in-order transmission and how to solve the cache miss problem with low memory overhead. Moreover, we can explore the interaction between load balancing and congestion control, and further study which flows need to be adjusted rate to alleviate congestion by injecting fewer in-flight packets, which flows only need to be switched paths to prevent queue buildup by distributing traffic, and which flows need to be rerouted and adjusted rate concurrently. Thus, it is necessary to develop solutions that leverage programmable data plane capabilities to intelligently route traffic, considering both congestion levels and path characteristics.

To ensure the validity and efficacy of the above proposed directions, simulations and testing in real-world scenarios are required to evaluate the impact on congestion alleviation, packet ordering, and overall network performance efficiency. Metrics such as packet delivery order, memory usage efficiency, and congestion mitigation effectiveness are key indicators.

Miscellaneous Issues

Background and Motivations

Nowadays, RDMA over Converged Ethernet with hop-by-hop PFC mechanism has been deployed in DCNs at scale by public cloud providers such as Microsoft, Alibaba and Google [2–4, 8, 10]. Although PFC effectively prevents packet loss due to buffer overflow, some serious phenomenons like deadlock and PFC storm occur concomitantly in the practical scenarios. Deadlock is caused by Cyclic Buffer Dependency (CBD), while all switches in this circle wait for each other to resume packet transmission [5, 9]. Once deadlock happens, it cannot break the interdependence loop spontaneously due to all ports in the CBD cease forever. Worse still, a global deadlock or PFC pause frame storm caused by congestion spreading will damage and shutdown the whole network. Due to the above serious problems occurred in the limited-scale networks, the enterprise datacenter operators cannot running RoCEv2 at large-scale DCNs.

To avoid fatal PFC's side-effects and enable large-scale RDMA networks, lossy RDMA DCNs have attracted significant attention recently [7, 15]. Without the guarantee of no packet loss, it is necessary to solve the problem of how to effectively retransmit caused by buffer overflow. Otherwise, a large number of inefficient retransmissions caused by inherent bursty traffic cannot enhance the scalability of RoCE. In addition, the connection scalability is also a crucial problem that the goodput collapses with the increase number of connections built on the RDMA NIC. As described in SRNIC [15], the aggregate throughput of RNICs collapses once the number of QPs exceeds 256. For example, the throughput of Mellanox CX-6 decreases 46% from 97Gbps to 52Gbps with the number of QPs increases from 128 to 16384. The root cause of the throughput performance degradation is the cache miss between the limited on-chip SRAM on the NIC and the host memory for context switching.

It is very challenging to take all these factors into consideration simultaneously to predict and prevent PFC triggering in distributed networks, and it is more difficult to avoid negative impacts of various PFC problems after PFC triggering.

Challenges

With the scale of datacenter network growing rapidly and the traffic increasing drastically, the datacenter operators cannot manipulate network configurations well manually to avoid deadlock and congestion spreading issues. Thus, these issues have attracted considerable attentions. However, PFC triggering involves many factors, including the switch buffer size, link bandwidth, number of priority queues, number of congested flows, flow sending rate, number of related ingress ports and PFC triggering threshold. It is very challenging to take all these factors into consideration simultaneously to predict and prevent PFC triggering in distributed networks, and it is more difficult to avoid negative impacts of various PFC problems after PFC triggering.

Traditional RoCEv2 transport performs inefficient go-back-N retransmission and only supports hundreds of connections. However, scalable RDMA transport in lossy RDMA network requires to be quite resilient to packet loss and massive connections. Based on the limited hardware resources on the RDMA NIC, the improved RoCE transport meets a series of new challenges, including redesigning reliable transmission algorithm, optimizing data structure and memory footprint, dealing with cache missing, etc. Moreover, following the trend of programmable transports, the advanced RoCE NIC requires to be programmable for each module in transport protocols to support flexible modification and reconfiguration of the data delivery algorithms such as congestion control and packet loss recovery mechanisms. Another key challenge is that the implementation of transport logic including packet generation and flow states query needs to satisfy the hardware timing constraints and consider the heterogeneity of the hardware infrastructures.

Existing Solutions

For Lossless RDMA Networks. To avoid the system traps in deadlock, GFC [9] and Tagger [5] are proposed to solve the cyclic buffer dependency problem by breaking the necessary conditions of deadlock and eliminating CBDs, respectively. Specifically, GFC [9] explores a new solution to break the hold and wait condition by manipulating the sending rate at a fine granularity. The flow control scheme ensures that the input rate at the ingress port matches the draining rate at the egress port and maintains the queue length in a steady state, so that all flows can continue to pass through the network even CBD exists. GFC's compatibility with mainstream lossless

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In navigating the challenges faced by distributed large-scale DCNs, a multifaceted approach is required to enhance the resilience, scalability, and performance of RDMA networks.

networks such as Converged Enhanced Ethernet and InfiniBand makes it applicable to RDMA networks. Tagger [5] reserves the expected lossless paths dynamically and allows the packets that not transmitted on lossless paths to be dropped under extreme congestion scenario, resulting in no CDB formation. The authors have theoretically and experimentally demonstrated that GFC and Tagger efficiently avoids deadlock and GFC introduces the average bandwidth consumption of less than 0.5%. TCD [12] designs a new congestion detection scheme to identify congested ports and flows accurately. TCD defines ternary states of switch ports, including congestion, non-congestion and undetermined states, and captures these states transition based on the queue length evolution. Due to the victim flows benefiting from TCD's ability to accurately detect congestion and undetermined states, the existing congestion control algorithms integrated with TCD can achieve up to 3.3 times and 2.0 times better median and 99th percentile FCT slowdown, respectively.

For Lossy RDMA Networks. To improve network scalability and connection scalability, SRNIC [15] devotes great efforts to designing a scalable RDMA NIC architecture. Guided by the insight that the on-chip data structures and their cache requirements in the RDMA NIC can be minimized by simultaneously optimizing protocol and architecture, SRNIC implements cache-free queue pairs scheduler and memory-free selective repeat to remove involved memory as many as possible. The experiments results show that SRNIC supports 10K performant connections on chip with high throughput and low latency. Compared with Mellanox RNICs, SRNIC achieves a line-rate throughput of up to 97 Gbps and extremely low 3μs latency. However, when the number of lost packets increases, the CPU overhead caused by software retransmission also increases.

Future Opportunities

In navigating the challenges faced by distributed large-scale DCNs, a multifaceted approach is required to enhance the resilience, scalability, and performance of RDMA networks. Through the analysis of existing network limitations and leveraging the potential of emerging technologies, three primary directions including hybrid RDMA network design, RoCE adaptation for lossy networks and domain-specific RDMA transports emerge as potential pathways to address the inherent issues within RDMA networks.

First, since the shutdown of the whole network is the most fatal damage for the distributed large-scale DCNs, one possible direction is to timely isolate the negative impacts of the hop-by-hop PFC mechanism in a limited scope of controllable local area networks. Although a series of serious problems such as deadlock and pausing storm caused by PFC triggering under the heavy congestion scenario, PFC can play a good role in a small range in terms of the advantages of simple deployment and effective preventing buffer overflow. Therefore, we can design a hybrid RDMA network by co-designing lossless and lossy networks divided from the perspective of datacenter topologies or switch priority queues. Moreover, it is a good opportunity to design machine learning based algorithms to dynamically tune the parameters in the partition mechanism to adapt to the network congestion changes. In this way, we can avoid the catastrophic effects of network-wide paralysis and continue to utilize the advantages of PFC to deliver reliable transmission. The improvement in transmission reliability and machine learning model effectiveness can be validated through simulation and real-world performance comparisons.

Second, the existing solutions for eliminating packet loss impose many restrictions on RDMA network configurations and harmful side-effects on performance. The other direction is to adapt RoCE to the lossy network to improve the scalability of RDMA network. With the help of relatively rich computing and storage resources in the advanced hardware such as FPGA, data processing unit (DPU) and Application Specific Integrated Circuit (ASIC), how to reasonably divide software and hardware functions for RoCE transports, including virtualization, tunneling, connection and buffer management, to combine the slow path and fast path for reliable data transmission directly affect the delay performance of RDMA. Complying with the trend of hardware offloading technique, how to implement a simple and efficient loss recovery mechanism to make RDMA network scalable and how to abstract the common logic modules to provide flexible RDMA transports programming on the high-speed RoCE NIC without hardware resources constraints are the important directions for further exploration and research. Evaluating scalability enhancements through simulations and hardware prototype testing can provide empirical evidence of success.

Third, in addition to the efforts on general RDMA transports, domain specific RDMA transports for diverse applications such as storage, distributed machine learning and high-performance computing are also the essential research directions. According to the traffic generation modes and traffic distribution characteristics of special applications, it is desirable to eliminate the transmission bottlenecks caused by the connection between heterogeneous hardware devices. Nowadays, there is a huge disparity in the development pace between network speed, computing capacity and cache size, so how to coordinate hardware resources and how to support direct network connectivity is a quite important issue in DCNs. However, only a few efforts have been made to allow the physical equipments such as FPGA accelerators and GPUs to directly control the RDMA NICs. Therefore, there is still a broad exploration and research space for tailoring RDMA transport protocols to the unique traffic generation and distribution characteristics of specific applications. To verify the effectiveness of the domain-specific transports, it is necessary to measure improvements in transmission efficiency and hardware resource coordination by implementing prototypes in relevant application environments.

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Conclusion

In this article, we made a survey on the current state-of-the-art of RDMA related work in Ethernet-based datacenter networks. We categorized the existing solutions into three categories, including single-path congestion control mechanisms, multi-path transports and miscellaneous issues, to show a general overview of this active research field common concerned by both academia and industry. We described the background and motivations, discussed the design challenges, reviewed the existing schemes and also pointed out the future research directions hoping to inspire new ways of thinking and make new contributions in this area.

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